CJ Petka

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EDUCATION

Northeastern University, Boston, MA **December 2029**

Candidate for Bachelor of Science in Mechanical Engineering and Computer Science — GPA: 4.00

Relevant Coursework: Statics, Thermodynamics, Physics for Engineering (1 & 2), Calculus 1–3, Differential Equations & Linear Algebra, Cornerstone of Engineering (1 & 2)

Awards: Ferretti Scholarship (2026), Dean's Scholarship, Dean's List (Fall 2025, Spring 2026)

Hempfield High School, Landisville, PA **May 2025**

Achievements: Class Rank 3/553 — GPA: 4.0

AP Courses: Calculus AB, Calculus BC, Physics C: Mechanics, Statistics, Computer Science A, Computer Science Principles, English Language and Composition, US Government and Politics, Comparative Government and Politics

Activities: Quizbowl, Dance Theatre, Esports, NHS

SKILLS

Programming and Tools: Python, C++, Java, C#, MicroPython, MATLAB, Git, Claude Code, Linux, multimeter

Electronics: Raspberry Pi Pico (RP2040, ARM Cortex-M0+), Arduino, I²C, UART, SPI, PWM, GPIO, circuit construction

CAD and Drafting: Autodesk Inventor, AutoCAD, SolidWorks, Onshape, Fusion 360, technical drawing

Fabrication: MIG/TIG welding, CNC mill, lathe, sheet metal, 3D printing (PLA, PETG, TPU), laser cutting, soldering

PROJECTS

React to the Pressure; Cornerstone of Engineering Team Project **Fall 2025**

- Drafted and laser cut the housing and designed 3D printed player controls for a pressure-based reaction-time game
- Wired and soldered the internal circuitry, then debugged the control software
- Demonstrated the finished game to the public at Northeastern's Engineering Expo

Steel Rail and Trolley Monitor Transport System; Self-developed Project **Spring 2024**

- Designed and fabricated a 13 ft steel rail and adjustable trolley to carry a VESA-mounted monitor across a fixed span
- Modeled the assembly and produced technical drawings in Autodesk Inventor
- Welded the structure across more than 100 welds and lathed custom wheels, then installed and tested.

Drone Package Delivery System; Self-developed Project **Spring 2025**

- Designed a servo-actuated release mechanism and airframe mount for a DJI Mini 2, sized to its payload margin
- Programmed two Raspberry Pi Picos in MicroPython to trigger the release over a UART radio link
- Tested the system with a live, beyond LOS drop of a small payload into a person's hand

WORK EXPERIENCE

Computer Systems Consultant — Freelance, Lancaster, PA **2021 – Present**

- Consult with clients on custom computer builds and specify components against performance needs and budget
- Assemble, configure, and test finished systems, then troubleshoot hardware and software for repeat clients

Retail Associate — S. Clyde Weaver, Lancaster, PA **2021 – Present**

- Run register and cash operations through peak-volume periods at a specialty food producer and market
- Handle food production and preparation alongside front-of-store service, and train new hires on register and production procedures

COMMUNITY SERVICE

Tae Kwon Do Instructor (ITF), IV Dan Black Belt **2013 – Present**

- Teach classes and mentor students; compete in patterns, sparring, and board breaking

A/V and Streaming Technician — Holy Trinity Lutheran Church, Lancaster, PA **2021 – Present**

- Run live audio and video for weekly streamed services and troubleshoot failures mid-broadcast
- Volunteer with a weekly meal program serving more than 150 people in Lancaster City

INTERESTS

Ballroom dance; 3D printing, welding, and metal fabrication